Model Selection

★ Occam's Razor :- Going for simplicity

Suppose there are 10 models → M_1, ----- , M_5

Let M_1 → didn't perform well ✗
M_2 → " " " ✗
M_3 → Performed well ✓
M_4 → " " ✓
M_5 → " " ✓

Out of M_3, M_4 & M_5, the model which is the simplest, we will select that model. This is known as Occam's Razor.

★ Model Complexity :-

↳ means how many features, parameters, layers, functions or calculations a model uses.

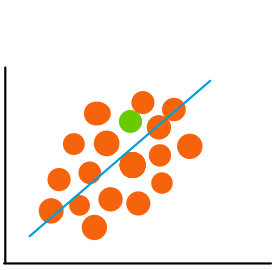
Complex Model	Simple Model
Complex Architecture	Simple Architecture
Millions of weights	few Weights
Overfitting	Perfectly fit / Underfitting
Resource Heavy	Resource Efficient
Ex- Bagging, Boosting, Deep learning models	Ex - Linear Regression, KNN

★ Overfitting & Underfitting :-

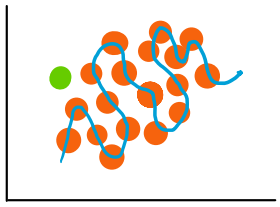
Underfitting :- Model does not perform well on train data and test data (poor model training)

data (poor model training)

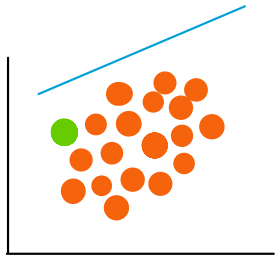
Overfitting :- Model performs exclemely well on training data & gives poor performance on test / unseen data. (very good model training)



Perfect fitting
or
Normal fitting



Overfitting



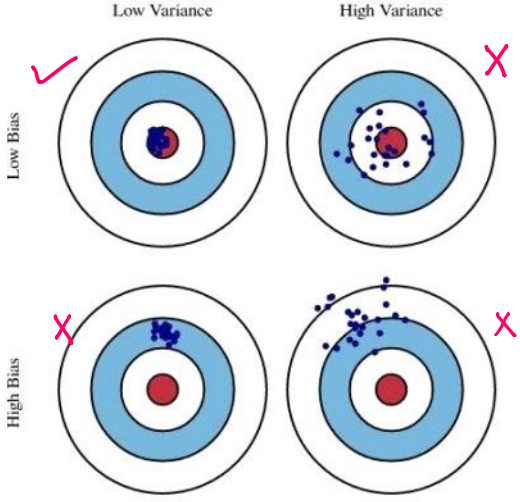
Underfitting

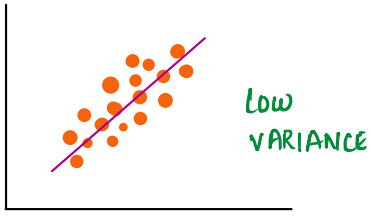
line drawn ⟶ Model is trained

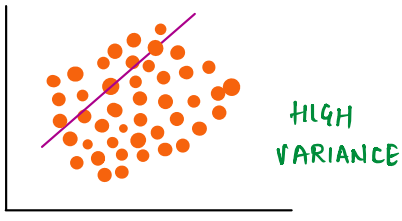
* Bias-Variance Trade-off :-

Bias → Error [diff b/w actual & predicted values]

Variance → Spread [diff b/w training & testi score]







Bias	Variance

Ex- Train score | Test Score

Bias	Variance			ex-	train score	test score
High	High	⇝	Underfitting		50%	10%
High	low		"		50%	48%
low	High	⇝	Overfitting		90%	72%
low	low	⇝	Normal fitting		92%	90%

A good model should have low bias & low variance.

	Train Data	Test Data
Underfitting	Poor	Poor
Overfitting	Best	Poor
Normal fitting	Best	Best

* Parametric & Non-Parametric Models :-

Parametric Models :- Uses specific mathematical form & can be described using fixed no. of parameters.

Ex- Linear Regression
Logistic Regression
Naive Bayes

Non-Parametric Models :- Does not use any specific mathematical form. Model complexity increases with multiple hyperparameters & increasing data

↓ increasing data

Ex- KNN

Decision Trees

Bagging

Boosting